

# Seminar 4: Discrete Random Variables

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# 1 Probability Distributions and Random Variables

In the previous lectures, probabilities were assigned directly to events in a probability space  $(\Omega, \mathcal{F}, P)$ . In practice, however, we are often interested in numerical quantities associated with the outcomes of an experiment, such as the number of heads obtained when tossing a coin several times, the number of customers arriving at a store during one hour, or the height of a randomly selected individual.

Such quantities are modeled by **random variables**. To study them, we first introduce probability distributions on the real line.

## 1.1 Borel $\sigma$ -Algebra

The **Borel  $\sigma$ -algebra** on  $\mathbb{R}$ , denoted by  $\mathcal{B}(\mathbb{R})$ , is the smallest  $\sigma$ -algebra containing all open subsets of  $\mathbb{R}$ .

Elements of  $\mathcal{B}(\mathbb{R})$  are called **Borel sets**. In particular, every open interval, closed interval, half-open interval, countable union, countable intersection, and complement of such sets belongs to  $\mathcal{B}(\mathbb{R})$ .

## 1.2 Measurable Functions and Random Variables

Let  $(\Omega, \mathcal{F})$  and  $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$  be measurable spaces. A function

$$\xi : \Omega \rightarrow \mathbb{R}$$

is called **measurable** if

$$\xi^{-1}(B) = \{\omega \in \Omega : \xi(\omega) \in B\} \in \mathcal{F} \quad \forall B \in \mathcal{B}(\mathbb{R}).$$

A **random variable** on a probability space  $(\Omega, \mathcal{F}, P)$  is simply a measurable function  $\xi : \Omega \rightarrow \mathbb{R}$ .

We denote by

$$X_\xi = \{\xi(\omega) : \omega \in \Omega\}$$

the set of all possible values of  $\xi$ .

If  $X_\xi$  is finite or countable, then  $\xi$  is called a **discrete random variable**.

### 1.3 Probability Distributions

A **probability distribution** is any probability measure on  $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ .

That is,

$$P : \mathcal{B}(\mathbb{R}) \rightarrow [0, 1].$$

If there exists a finite or countable set  $X \subseteq \mathbb{R}$  such that  $P(X) = 1$ , then  $P$  is called a **discrete probability distribution**.

### 1.4 Distribution of a Random Variable

Let  $\xi$  be a random variable on  $(\Omega, \mathcal{F}, P)$ .

The **probability distribution of  $\xi$**  is the probability measure  $P_\xi$  on  $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$  defined by

$$P_\xi(B) = P(\{\omega \in \Omega : \xi(\omega) \in B\}) = P(\xi \in B), \quad B \in \mathcal{B}(\mathbb{R}).$$

Thus, the distribution of a random variable completely describes the probabilities of all events involving its values.

### 1.5 Probability Mass Function (PMF)

Let  $\xi$  be a discrete random variable and let  $X_\xi = \{x_1, x_2, \dots\}$ .

The **probability mass function (PMF)** of  $\xi$  is the function

$$p_\xi : \mathbb{R} \rightarrow [0, 1]$$

defined by

$$p_\xi(x_k) = P(\xi = x_k), \quad x_k \in X_\xi,$$

and  $p_{\xi}(x) = 0$ ,  $x \notin X_{\xi}$

The PMF completely determines the distribution of a discrete random variable and satisfies  $\sum_{x \in X_{\xi}} p_{\xi}(x) = 1$ .

## 2 Examples of Discrete Probability Distributions

### 2.1 Uniform Distribution

Let  $X$  be a finite set. The uniform distribution on  $X$  is defined by

$$P(x) = \frac{1}{|X|}, \quad x \in X.$$

### 2.2 Bernoulli Distribution

For  $p \in (0, 1)$ ,  $X = \{0, 1\}$ , and

$$P(0) = 1 - p, \quad P(1) = p.$$

We write

$$\xi \sim \text{Ber}(p).$$

### 2.3 Binomial Distribution

For  $n \in \mathbb{N}$  and  $p \in (0, 1)$ ,  $X = \{0, 1, \dots, n\}$ , and

$$P(k) = \binom{n}{k} p^k (1 - p)^{n-k}.$$

We write

$$\xi \sim \text{Bin}(n, p).$$

## 2.4 Geometric Distribution

For  $p \in (0, 1)$ ,  $X = \mathbb{N} \setminus \{0\}$ , and

$$P(k) = (1 - p)^{k-1}p, \quad k \geq 1.$$

We write

$$\xi \sim \text{Geom}(p).$$

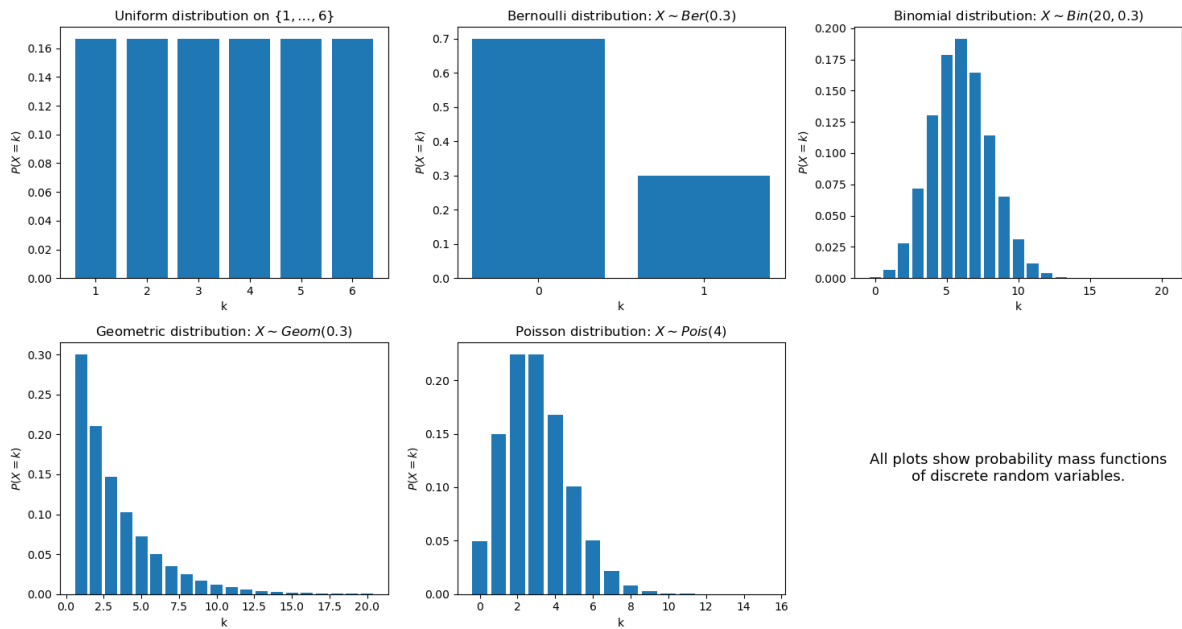
## 2.5 Poisson Distribution

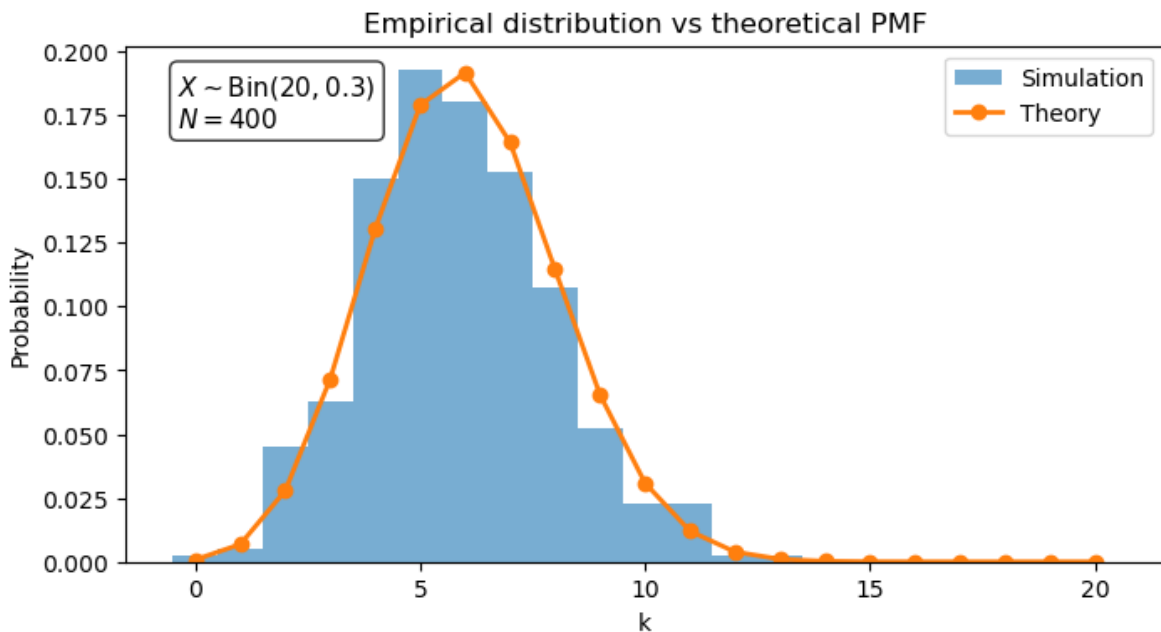
For  $\lambda > 0$ ,  $X = \{0, 1, 2, \dots\}$ , and

$$P(k) = \frac{\lambda^k e^{-\lambda}}{k!}.$$

We write

$$\xi \sim \text{Pois}(\lambda).$$





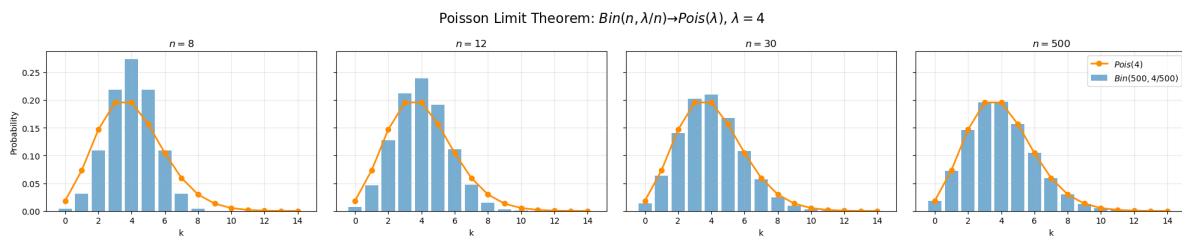
### 3 Poisson Limit Theorem

Let  $\xi_n \sim \text{Bin}(n, p_n)$  and suppose that  $np_n \rightarrow \lambda > 0$ .

Then for every fixed  $k \geq 0$ ,

$$P(\xi_n = k) \rightarrow \frac{\lambda^k e^{-\lambda}}{k!}.$$

In other words, the binomial distribution converges to the Poisson distribution when the number of trials becomes large and the success probability becomes small in such a way that the expected number of successes remains approximately constant.



## 4 Independence of Random Variables

Two discrete random variables  $\xi$  and  $\eta$  are called **independent** if

$$P(\xi = x, \eta = y) = P(\xi = x)P(\eta = y)$$

for all possible values  $x \in X_\xi$  and  $y \in X_\eta$ .

Similarly, a collection of random variables  $\xi_1, \dots, \xi_n$  is called mutually independent if

$$P(\xi_{i_1} = x_{i_1}, \dots, \xi_{i_k} = x_{i_k}) = \prod_{j=1}^k P(\xi_{i_j} = x_{i_j})$$

for every subset  $\{i_1, \dots, i_k\} \subseteq \{1, \dots, n\}$ .

### 4.1 Convolution of Discrete Distributions

Let  $\xi$  and  $\eta$  be independent discrete random variables. Then

$$P(\xi + \eta = k) = \sum_{i=-\infty}^{+\infty} P(\xi = i)P(\eta = k - i).$$

This formula is called the **convolution formula**.

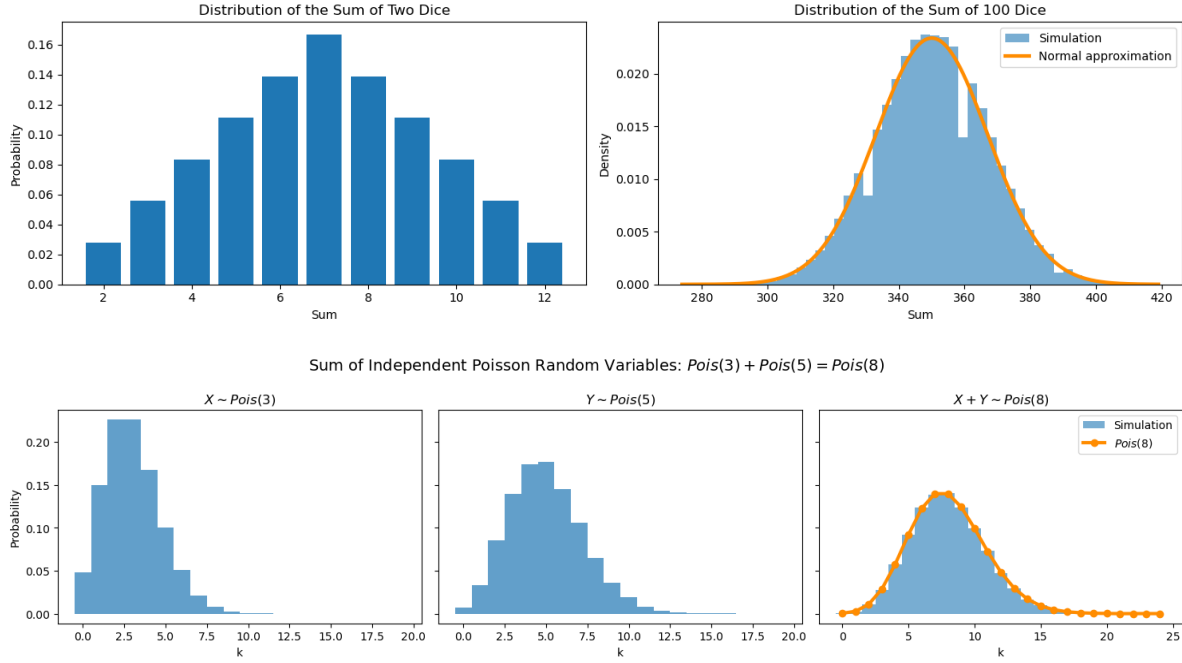
Important examples:

- If  $\xi \sim \text{Bin}(n, p)$  and  $\eta \sim \text{Bin}(m, p)$  are independent, then

$$\xi + \eta \sim \text{Bin}(n + m, p).$$

- If  $\xi \sim \text{Pois}(\lambda_1)$  and  $\eta \sim \text{Pois}(\lambda_2)$  are independent, then

$$\xi + \eta \sim \text{Pois}(\lambda_1 + \lambda_2).$$



## 5 Discrete Random Vectors

So far we have studied a single random variable  $\xi : \Omega \rightarrow \mathbb{R}$ . In many applications, however, several random quantities are observed simultaneously. This leads to the notion of a random vector.

### 5.1 Definition

A **2-dimensional discrete random vector** is a pair of discrete random variables

$$(\xi, \eta).$$

More generally,

$$(\xi_1, \xi_2, \dots, \xi_n).$$

For every outcome  $\omega \in \Omega$ , the random vector assigns a point

$$(\xi_1(\omega), \xi_2(\omega), \dots, \xi_n(\omega)) \in \mathbb{R}^n.$$



The set of all possible values is called the **support** of the random vector.

## 5.2 Joint Probability Mass Function

The probability distribution of a discrete random vector is described by its **joint probability mass function (joint PMF)**.

### 5.2.1 Definition

For a discrete random vector  $(\xi, \eta)$ ,

$$p_{\xi, \eta}(x, y) = \mathbb{P}(\xi = x, \eta = y).$$

The value  $p_{\xi, \eta}(x, y)$  is the probability that both events occur simultaneously.

The joint PMF satisfies  $p_{\xi, \eta}(x, y) \geq 0$ , and

$$\sum_x \sum_y p_{\xi, \eta}(x, y) = 1.$$

## 5.3 Joint PMF Tables

A convenient way to represent a joint PMF is by a table.

### 5.3.1 Example: Two Fair Dice

For two independent fair dice,

$$p_{\xi, \eta}(x, y) = \frac{1}{36}, \quad x, y \in \{1, \dots, 6\}.$$

| $\xi \backslash \eta$ | 1              | 2              | 3              | 4              | 5              | 6              |
|-----------------------|----------------|----------------|----------------|----------------|----------------|----------------|
| 1                     | $\frac{1}{36}$ | $\frac{1}{36}$ | $\frac{1}{36}$ | $\frac{1}{36}$ | $\frac{1}{36}$ | $\frac{1}{36}$ |
| 2                     | $\frac{1}{36}$ | $\frac{1}{36}$ | $\frac{1}{36}$ | $\frac{1}{36}$ | $\frac{1}{36}$ | $\frac{1}{36}$ |
| 3                     | $\frac{1}{36}$ | $\frac{1}{36}$ | $\frac{1}{36}$ | $\frac{1}{36}$ | $\frac{1}{36}$ | $\frac{1}{36}$ |
| 4                     | $\frac{1}{36}$ | $\frac{1}{36}$ | $\frac{1}{36}$ | $\frac{1}{36}$ | $\frac{1}{36}$ | $\frac{1}{36}$ |
| 5                     | $\frac{1}{36}$ | $\frac{1}{36}$ | $\frac{1}{36}$ | $\frac{1}{36}$ | $\frac{1}{36}$ | $\frac{1}{36}$ |
| 6                     | $\frac{1}{36}$ | $\frac{1}{36}$ | $\frac{1}{36}$ | $\frac{1}{36}$ | $\frac{1}{36}$ | $\frac{1}{36}$ |

Every cell corresponds to one outcome  $(x, y)$ .

The sum of all entries of the table must be equal to 1.

## 6 Marginal Distributions

The distributions of  $\xi$  and  $\eta$  can be recovered from the joint PMF.

### 6.1 Definition

The **marginal PMF** of  $\xi$  is

$$p_{\xi}(x) = \sum_y p_{\xi,\eta}(x, y).$$

Similarly,

$$p_{\eta}(y) = \sum_x p_{\xi,\eta}(x, y).$$

In words, we sum over all possible values of the other variable.

#### 6.1.1 Example

For two fair dice,

$$p_{\xi}(x) = \sum_{y=1}^6 \frac{1}{36} = \frac{1}{6}.$$

## 6.2 Independence

The notion of independence extends naturally to random vectors.

### 6.2.1 Definition

Random variables  $\xi$  and  $\eta$  are **independent** if

$$p_{\xi,\eta}(x,y) = p_{\xi}(x)p_{\eta}(y) \quad \forall x,y.$$

Thus, the probability of a pair of outcomes factors into the product of the marginal probabilities.

## 7 Discrete distributions in ML

### 7.1 Logistic Regression

Many machine learning problems involve predicting a binary outcome.

Suppose

$$Y = \begin{cases} 1, & \text{success,} \\ 0, & \text{failure.} \end{cases}$$

Then

$$Y \sim \text{Ber}(p),$$

where  $p$  is the probability of success. Logistic regression is used when the response variable has only two possible outcomes.

Examples include:

- Will a customer purchase a product?
- Will a student pass an exam?
- Will a patient develop a disease?
- Is an email spam or not spam?
- Will a user click on an advertisement?

In logistic regression, this probability depends on explanatory variables. For example, an online store may use

$$(x_1, x_2, x_3) = (\text{age, income, number of previous purchases})$$

to predict the probability that a customer makes a purchase.

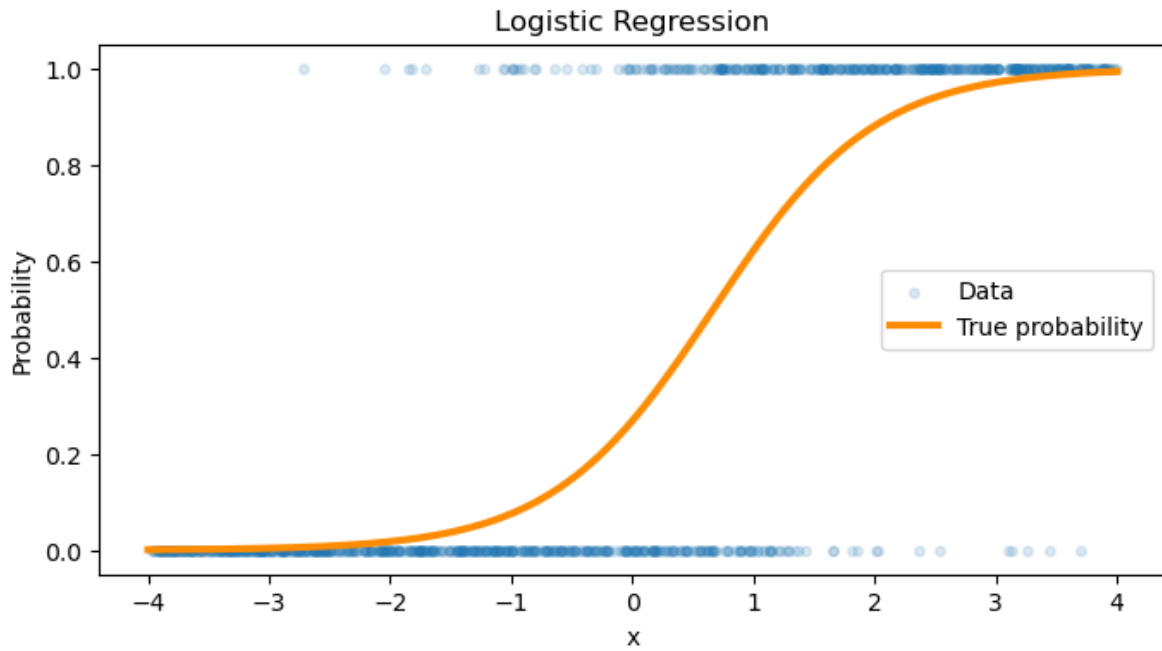
For a single feature  $x$ , the model assumes

$$p(x) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x)}}.$$

Thus,

$$Y \mid x \sim \text{Ber}(p(x)).$$

Logistic regression is one of the most widely used classification methods in machine learning.



## 7.2 Poisson Regression

Many applications involve predicting counts.

Examples include:

- number of website visits;
- number of accidents;
- number of insurance claims;
- number of emails received.

In such situations, it is natural to assume

$$Y \sim \text{Pois}(\lambda),$$

where  $\lambda$  is the expected count.

Poisson regression models the parameter  $\lambda$  as

$$\lambda(x) = e^{\beta_0 + \beta_1 x}.$$

The exponential function guarantees that  $\lambda(x) > 0$ .

For example, a company may use

$$(x_1, x_2, x_3) = (\text{advertising budget}, \text{season}, \text{number of users})$$

to predict the expected number of website visits. Thus,

$$Y \mid x \sim \text{Pois}(\lambda(x)).$$

Poisson regression is one of the most important generalized linear models.

